

Task 2: Clear Away Normalisation

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UNF

CONTRACT_WASTE (contract_no, waste_type_id, waste_type_desc, ctwaste_frequency, (collrun_date, truck_vin, truck_rego, truck_make, truck_model, truck_year, (collbin_rfid, collbin_kgs, collbin_overweight)))

1NF

CONTRACT_WASTE(contract_no, waste_type_id, waste_type_desc, ctwaste_frequency)

CK:

contract_no, waste_type_id

COLLECTION_RUN(contract_no, waste_type_id, collrun_date, truck_vin, truck_rego, truck_make, truck_model, truck_year)

CK:

contract_no, waste_type_id, collrun_date

COLLECTION_BIN(contract_no, waste_type_id, collrun_date, collbin_rfid, collbin_kgs, collbin_overweight)

CK:

contract_no, waste_type_id, collrun_date, collbin_rfid
collbin_rfid, collrun_date

Partial dependency:

waste_type_id -> waste_type_desc

2NF

CONTRACT_WASTE(contract_no, waste_type_id, ctwaste_frequency)

WASTE_TYPE(waste_type_id, waste_type_desc)

COLLECTION_RUN(contract_no, waste_type_id, collrun_date, truck_vin, truck_rego, truck_make, truck_model, truck_year)

COLLECTION_BIN(contract_no, waste_type_id, collrun_date, collbin_rfid, collbin_kgs, collbin_overweight)

Transitive dependency:

truck_vin -> truck_make, truck_model, truck_year

3NF

CONTRACT_WASTE(contract_no, waste_type_id, ctwaste_frequency)

WASTE_TYPE(waste_type_id, waste_type_desc)

COLLECTION_RUN(contract_no, waste_type_id, collrun_date, truck_vin, truck_rego)

TRUCK(truck_vin, truck_make, truck_model, truck_year)

COLLECTION_BIN(contract_no, waste_type_id, collrun_date, collbin_rfid, collbin_kgs, collbin_overweight)

Full dependency:

contract_no, waste_type_id -> ctwaste_frequency

waste_type_id -> waste_type_desc

contract_no, waste_type_id, collrun_date -> truck_vin, truck_rego

truck_vin -> truck_make, truck_model, truck_year

contract_no, waste_type_id, collrun_date, collbin_rfid -> collbin_kgs, collbin_overweight